LUMEXA
Space-themed coding education for K–12

AGES 12–18 | BASIC PYTHON NEEDED

CURRICULUM OVERVIEW

Data Scientist Path

Build 3 Data Projects

A complete, project-driven journey from messy raw data to real, working predictive models. Students don't just learn about data science — they clean real datasets, build compelling visualizations, and train machine learning models that make real decisions.



3 REAL DATA PROJECTS
Analysis, dashboards and models

4 TESTS ALONG THE WAY
3 course tests + 1 final test

3	24	28	12–18
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
For learners who already know basic Python

BEFORE YOU START
Basic Python knowledge needed

LEARNING PROGRESSION

01 Python for Data
Load, clean and analyse real datasets with Pandas.

02 Data Visualisation
Charts and dashboards that tell a clear story.

03 Machine Learning Projects
Train, evaluate and deploy models that predict.

Learn by analysing, not by watching
Every lesson pairs a real concept with a real dataset — no toy examples, no invented numbers.

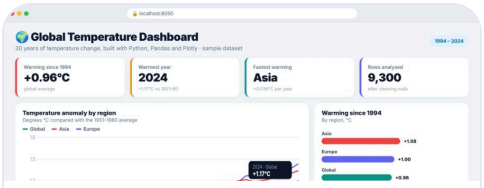
A portfolio, not just a certificate
Students finish with three real data projects they can present, explain and keep building on.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

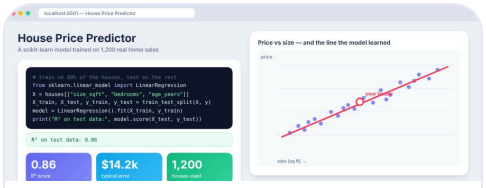
SAMPLE HANDS-ON PROJECTS STUDENTS BUILD



01 Sports Stats Analyser
Python · Pandas



02 Climate Change Dashboard
Python · Plotly · Dash



03 House Price Predictor
Python · scikit-learn

COURSE 01 · LESSONS 1–8

01 Python for Data

Load, clean, and analyse real-world datasets using Python

Python 3 | Pandas | NumPy | Jupyter Notebooks

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Every data project starts with messy, real-world data. Your child learns to wrangle it into something useful — finding patterns, calculating stats, and drawing conclusions.

PORTFOLIO PROJECTS

- Sports performance analysis
- School grade analysis
- City population trends

WHY IT MATTERS

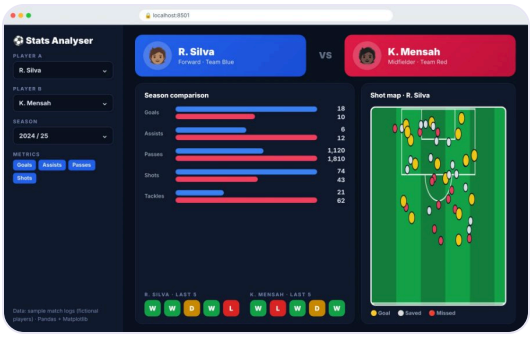
Real data is never clean. Learning to load, filter, and fix it in Pandas is the single most-used skill in data science — the foundation every chart, model, and insight in this path depends on.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. Why Data Science Matters (Real Examples)
2. Python Refresher and Jupyter Setup
3. Loading Data from CSV and Excel
4. Pandas — Selecting, Filtering, Sorting
5. Cleaning Data — Nulls, Duplicates, Types
6. Aggregations — Group By, Pivot Tables
7. Statistical Summaries and Correlation
8. Full Analysis Project with a Real Dataset

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Sports Stats Analyser

Python · Pandas

A player-comparison tool built from a real season of sports data: loaded, cleaned, grouped and compared with Pandas.

COURSE TEST

Python for Data — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 02 · LESSONS 9–16

02 Data Visualisation

Turn raw data into compelling, interactive charts and dashboards

Python | Matplotlib | Seaborn | Plotly | Dash

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Data without visuals is just numbers. Your child learns to create charts that tell stories — the kind that make people say "I never knew that" when they look at the data.

PORTFOLIO PROJECTS

- Climate change dashboard
- Market trends chart
- Sports comparison visualiser

WHY IT MATTERS

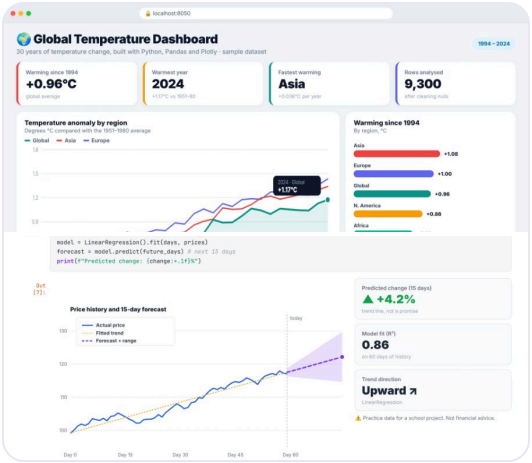
A brilliant analysis nobody can read is wasted. This course teaches students to choose the right chart, design it clearly, and build interactive dashboards — the way data scientists actually communicate results.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. The science of good data visualisation
2. Matplotlib: line, bar, scatter plots
3. Seaborn: statistical visualisations
4. Choosing the right chart for your data
5. Color, labels, and storytelling
6. Interactive charts with Plotly
7. Building a Dash dashboard
8. Final: your own interactive dashboard

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECTS

Climate Change Dashboard · Market Trends Forecast

Python · Plotly · Dash | Python · Matplotlib

Charts that tell a story: an interactive climate dashboard and a market trend chart with a forecast line.

COURSE TEST

Data Visualisation — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 03 · LESSONS 17–24

03 Machine Learning Projects

Build predictive models that make real decisions from real data

Python · scikit-learn · XGBoost · Joblib

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

The pinnacle of data science: teaching a computer to predict outcomes. Your child builds real machine learning models — the same technology behind Netflix recommendations and fraud detection.

PORTFOLIO PROJECTS

- House price predictor
- Customer churn predictor
- Movie recommendation engine

WHY IT MATTERS

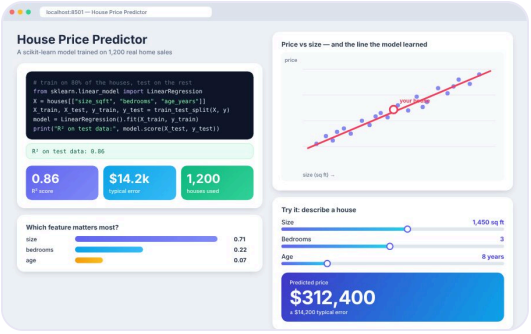
This is where cleaned data and clear charts become real predictions. Students learn the complete machine learning workflow — training, evaluating, tuning, and deploying a model — the same process used in industry.

8 LESSONS | **3** PROJECTS | **1** COURSE TEST

LESSON BREAKDOWN

1. How machine learning works (no maths jargon)
2. Types of ML: regression vs classification
3. Train/test splits and model evaluation
4. Linear regression from scratch
5. Decision trees and random forests
6. Feature engineering and selection
7. Model tuning and improving accuracy
8. Deploying a prediction model as an API

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

House Price Predictor

Python · scikit-learn

A model trained on real housing data that predicts a price from size, rooms and age — with its accuracy shown honestly.

COURSE TEST

Machine Learning Projects — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

HOW STUDENTS ARE TESTED · 4 TESTS

Level-Up Checkpoints

Data science is only as good as the thinking behind it. Course tests check that students can explain how they cleaned, charted and modelled — and a final test closes the path.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL



COURSE TEST · every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

- **30** multiple-choice questions — choose one of four answers
- **10** true / false questions
- **40** marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

One final test across data analysis, visualisation and machine learning: questions plus a small practical project.

- **25** multiple-choice questions
- **5** true / false questions
- **20** marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Which Pandas method gives the average of a column?

- A .sort()
- B .drop()
- C .mean() ✓ CORRECT
- D .head()

SAMPLE · TRUE / FALSE

Duplicate rows never affect an analysis.

True | False ✓

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 3 courses, 24 lessons and 4 tests, students progress from loading a messy spreadsheet to training a deployable prediction model — walking away with three real, working data projects for their portfolio.

3 COURSES · 24 LESSONS

01 Python for Data
Sports performance analysis · School grade analysis · City population trends
8 lessons · + course test

02 Data Visualisation
Climate change dashboard · Market trends chart · Sports comparison visualiser
8 lessons · + course test

03 Machine Learning Projects
House price predictor · Customer churn predictor · Movie recommendation engine
8 lessons · + course test

HOW STUDENTS ARE TESTED · 4 TESTS

COURSE TEST 1
Python for Data

COURSE TEST 2
Data Visualisation

COURSE TEST 3
Machine Learning Projects

FINAL TEST
One final test on the whole curriculum
One final test across data analysis, visualisation and machine learning: questions plus a small practical project.

BY THE END, STUDENTS CAN

- Load, clean, and analyse a real-world dataset using Python and Pandas.
- Design clear, compelling charts and build an interactive data dashboard.
- Train, evaluate, and deploy a real machine learning prediction model.

Ready to start analysing?
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa Data Scientist · Ages 12–18